

Roman grism line-grid simulation package

Wave 1b — emission-line point sources with exact truth tables, for line-by-line validation of grism dispersion and extraction

DRAFT · 2026-07-24

roman_disperser · branch feature/optical-model-line-test · commit eee413e

Where the data is

Run store (Roman AWS cluster; readable from head and compute nodes):

/mnt/roman-science/grs/line-tests-20260724/

— catalog_lines/	408K	input catalog: positions + line SEDs + line list
— lines_stpsf/	4.5G	simulations, real (STPSF) PSFs – one dir per PA
— lines_gauss/	4.5G	simulations, Gaussian-control PSFs – same layout
— lines_stpsf_l2/	7.3G	romanisim L2 ASDF products for lines_stpsf
— lines_gauss_l2/	7.3G	same for lines_gauss
— scripts/	36K	verbatim campaign drivers + configs + pointings
— slurm-meta/	148K	per-job audit: task script + environment record
— logs/	2.5M	per-file romanisim wrap logs

Pointing directories are named pointing_{pa000,pa010}_001.001.001.001.001.001 (suffix = APT plan.pass.segment.observation.visit.exposure).

Code: roman_disperser repo, branch feature/optical-model-line-test (PR #17).

What was simulated — sources and spectra

15,112 point sources: every star with $F158 \leq 20$ in the parent reference catalog.

All stars carry the same spectrum: five Gaussian emission lines on a flat continuum that is then removed — the images contain lines only.

Continuum anchor: mag 16.3072 in F158 (flux_scale = 3.0000e-7 maggies); line amplitudes are defined relative to that (removed) continuum.

line_id	λ [Å, vacuum]	FWHM [Å]	amp / continuum	line flux [erg s ⁻¹ cm ⁻²]	detected e ⁻ (+1 order, 190.22 s)
0	11000	10	5.00	1.44e-14	22,600
1	12500	10	6.25	1.39e-14	32,000
2	15500	10	7.50	1.09e-14	27,500
3	17000	10	8.75	1.05e-14	24,800
4	19000	10	10.00	9.63e-15	19,300

Detected e⁻ = 33×33 px aperture sum on the noise-free image × exposure time.

What was simulated — run matrix and observation

Four runs = {STPSF PSFs, Gaussian PSFs} $\times$ {PA 0°, PA 10°}, all 18 SCAs each:

STPSF: real chromatic, field-dependent, asymmetric PSFs (4×4 spatial grid $\times$ 56 wavelengths per SCA, 4× oversampled)

Gaussian: centred symmetric Gaussians (FWHM 2.5 px) in the same cache format — a PSF-shape control with identical geometry

PA 10°: the same pointing rolled by 10° — a sky-orientation control

One pointing: RA 10.0°, Dec 0.0°, exposure 190.22 s, MA table 1036 (GRISM, 190.22287 s).

Spectral orders 0, +1, +2 are dispersed for every source.

All randomness is seeded (config seed 42); per-SCA RNG keys are recorded in the FITS headers and meta files.

What was simulated — pipeline

1. Simulate. JAX-based disperser renders all sources, orders and SCAs.

Output per SCA: FITS with a noise-free MODEL extension (counts/s) and a Poisson-sampled ISIM extension (counts at 190.22 s).

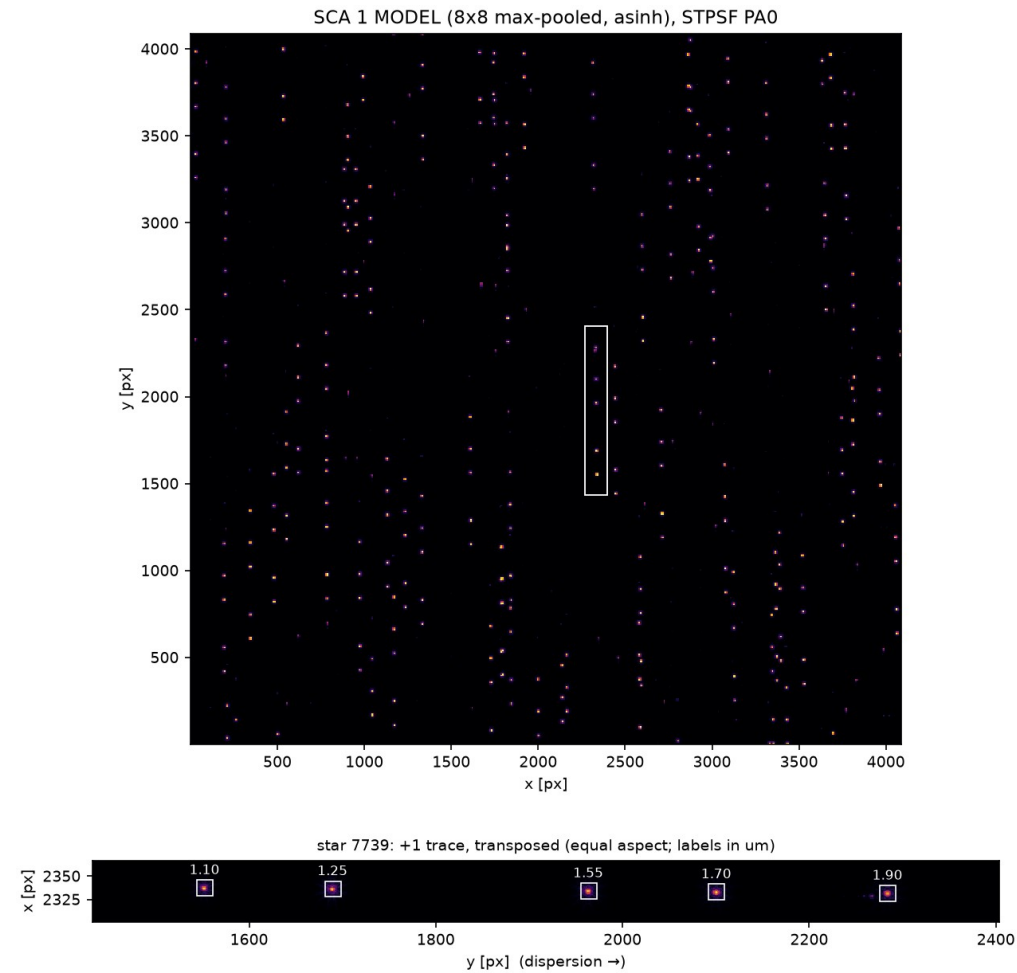
2. Wrap. romanisim converts each MODEL/ISIM to a Roman L2 ASDF (ramp simulation with MA table 1036, then ramp fit).

Output per SCA: *_l2.asdf with data / err / dq (DN/s), 11 resultants, effective exposure 190.22287 s.

3. Measure + truth. Every first-order (star, line) position is predicted by the optical model and measured on the noise-free MODEL image; independently, float64 truth positions are tabulated for all orders.

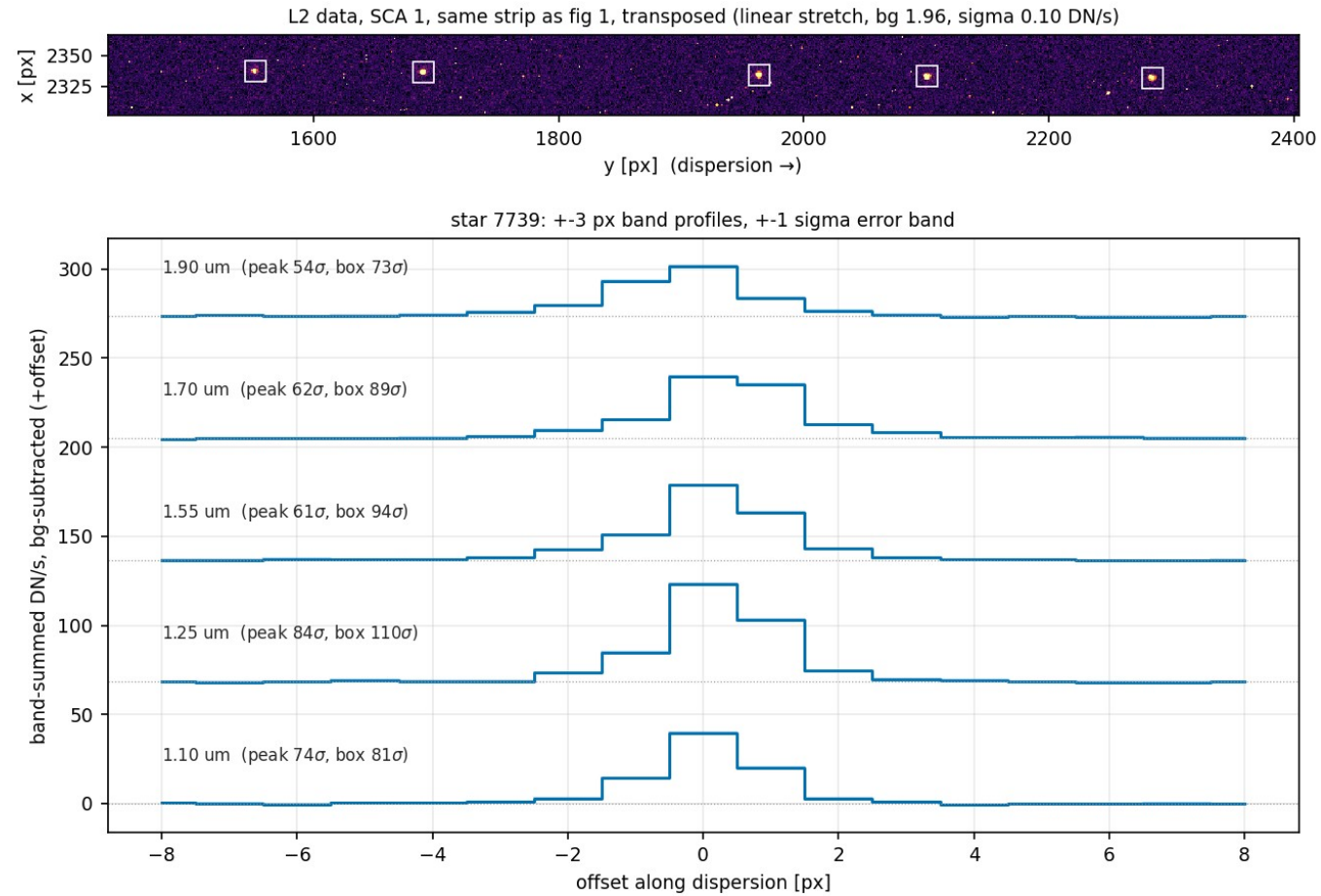
Output per pointing: residuals.parquet (closure measurement) and truth_lines.parquet (shipped truth table).

Simulated image (SCA 1, STPSF, PA 0)



Noise-free MODEL image. Top: full SCA (8×8 max-pooled for display). Bottom: one star's +1-order trace, transposed to equal aspect — five isolated emission-line spots with their 17×17 px measurement boxes.

L2 product and noise (same strip)



L2 data for the same strip (background 1.96 DN/s, robust σ 0.105 DN/s). Per-line peak SNR 54-84, box SNR 73-110; median peak SNR over all 322 measured boxes on SCA 1 is 66.

Line-position measurement

For every first-order (star, line) pair, on the noise-free MODEL image:

- Predict the line's detector position with the optical model (two independent implementations; they agree to ≤ 0.005 px).

- Cut a 17×17 px box at the prediction; drop the pair if the box is not fully on-detector (this is the only cut).

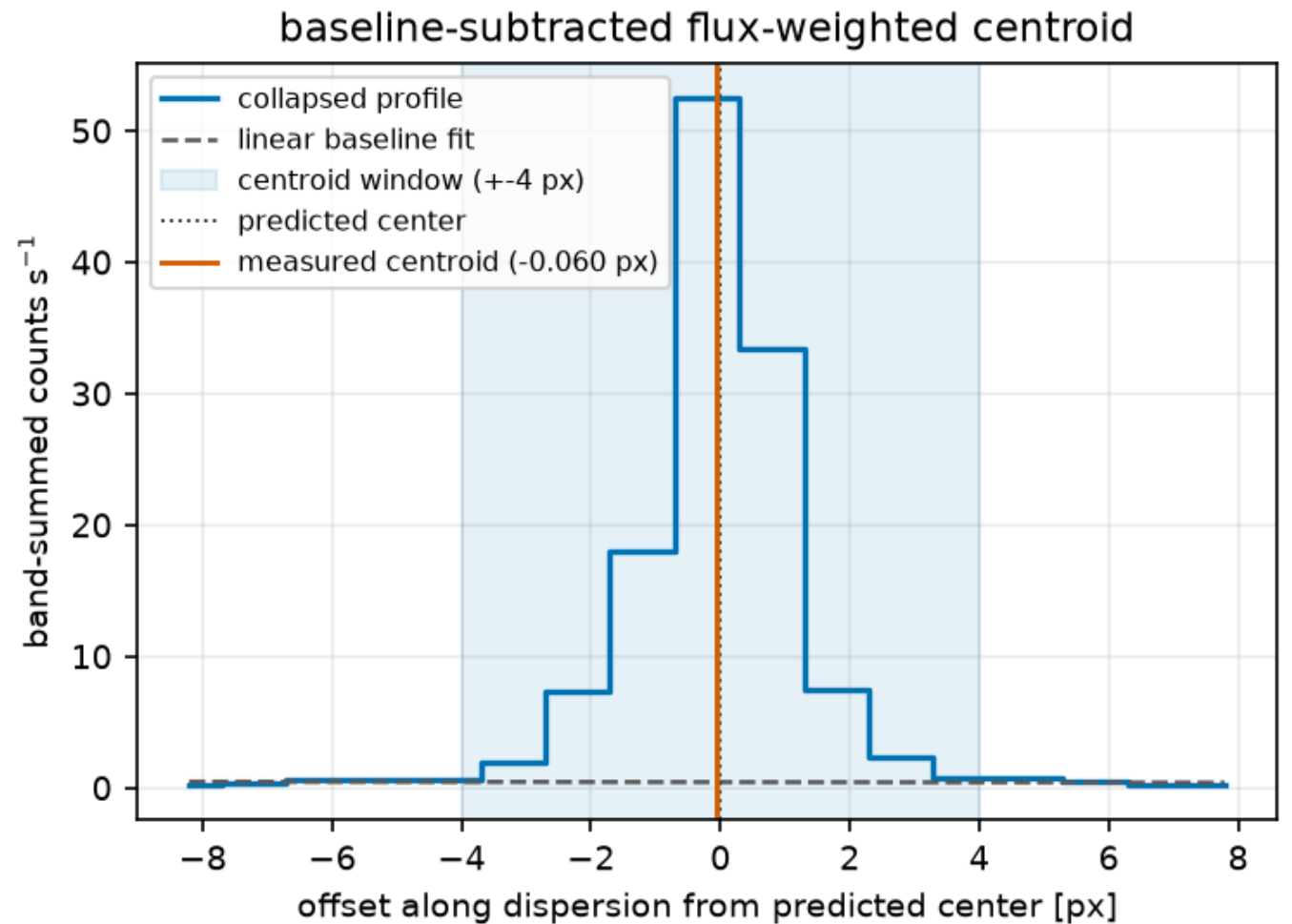
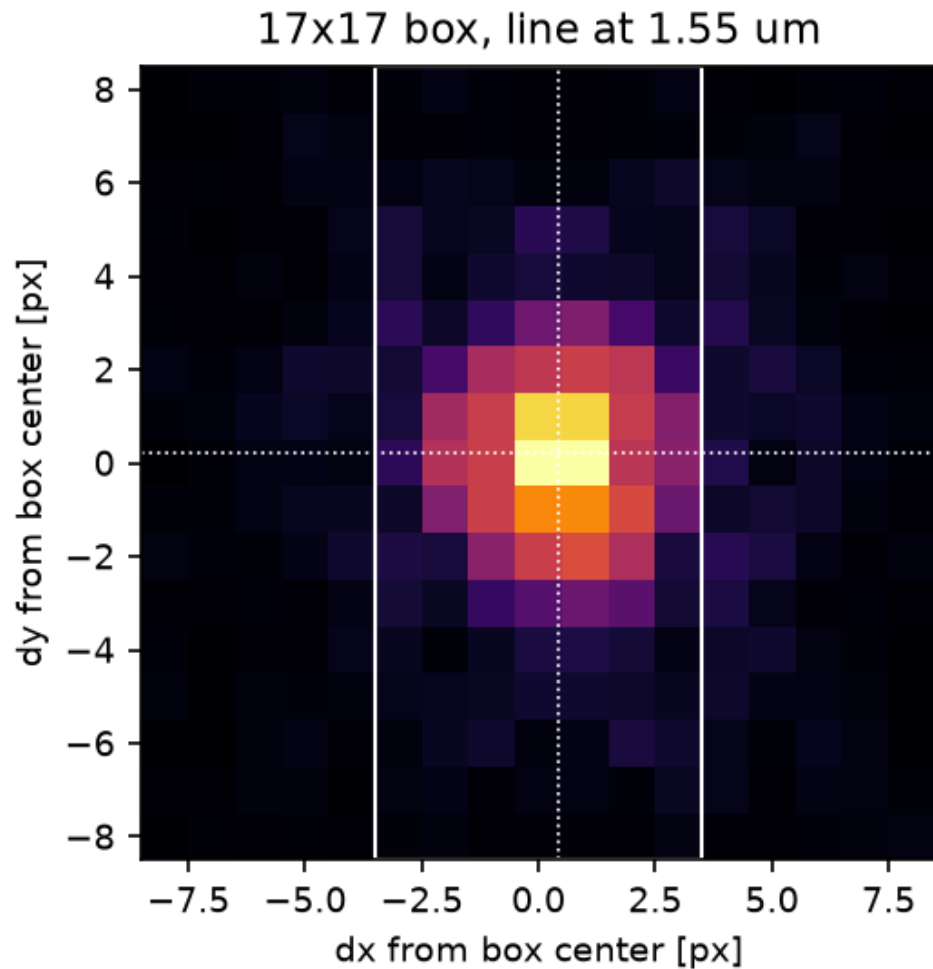
- Collapse a ± 3 px band across dispersion; fit and subtract a linear baseline to the off-line pixels ($|\Delta| > 4$ px).

- Measure the flux-weighted centroid within ± 4 px of the prediction.

Residuals $d_{\text{disp}} / d_{\text{cross}}$ are measured – predicted, projected on the local dispersion / cross-dispersion directions, in native pixels.

Convention: a “measured position” is a flux-weighted centroid. For the STPSF PSFs this differs from the PSF peak by up to ~ 0.08 px, and depends on the window (± 4 px is core-weighted; wing-weighted apertures move per-SCA offsets by up to $\sim 30\%$).

The estimator on one line



One 17×17 box (1.55 μm line, SCA 1): the ± 3 px collapse band, the band-summed profile, linear baseline, ± 4 px centroid window, and measured centroid vs prediction.

Results — summary

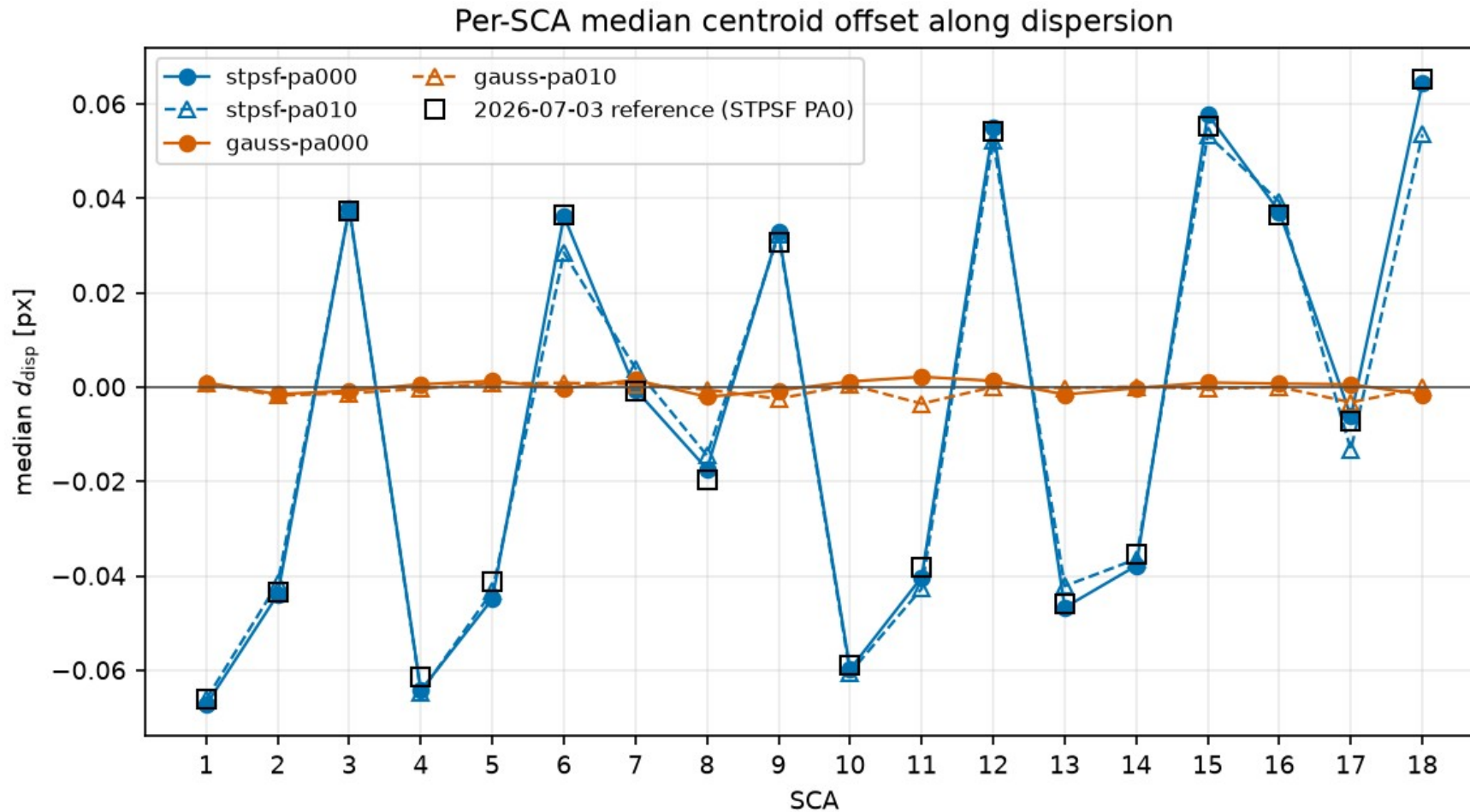
Statistics over all finite-centroid first-order boxes (MAD = unscaled median absolute deviation):

run		boxes (measured/total)	d_disp median [px]	d_disp MAD [px]	d_cross median [px]	optical-model implementations agree to [px]
stpsf	pa000	5290 / 8575	−0.015	0.041	+0.024	0.0040
stpsf	pa010	5266 / 8465	−0.017	0.040	+0.025	0.0053
gauss	pa000	5290 / 8575	+0.000	0.010	−0.001	0.0040
gauss	pa010	5266 / 8465	−0.000	0.010	−0.001	0.0053

Unmeasured boxes are all geometric edge drops (17×17 box not fully on-detector); none are dropped for lack of flux.

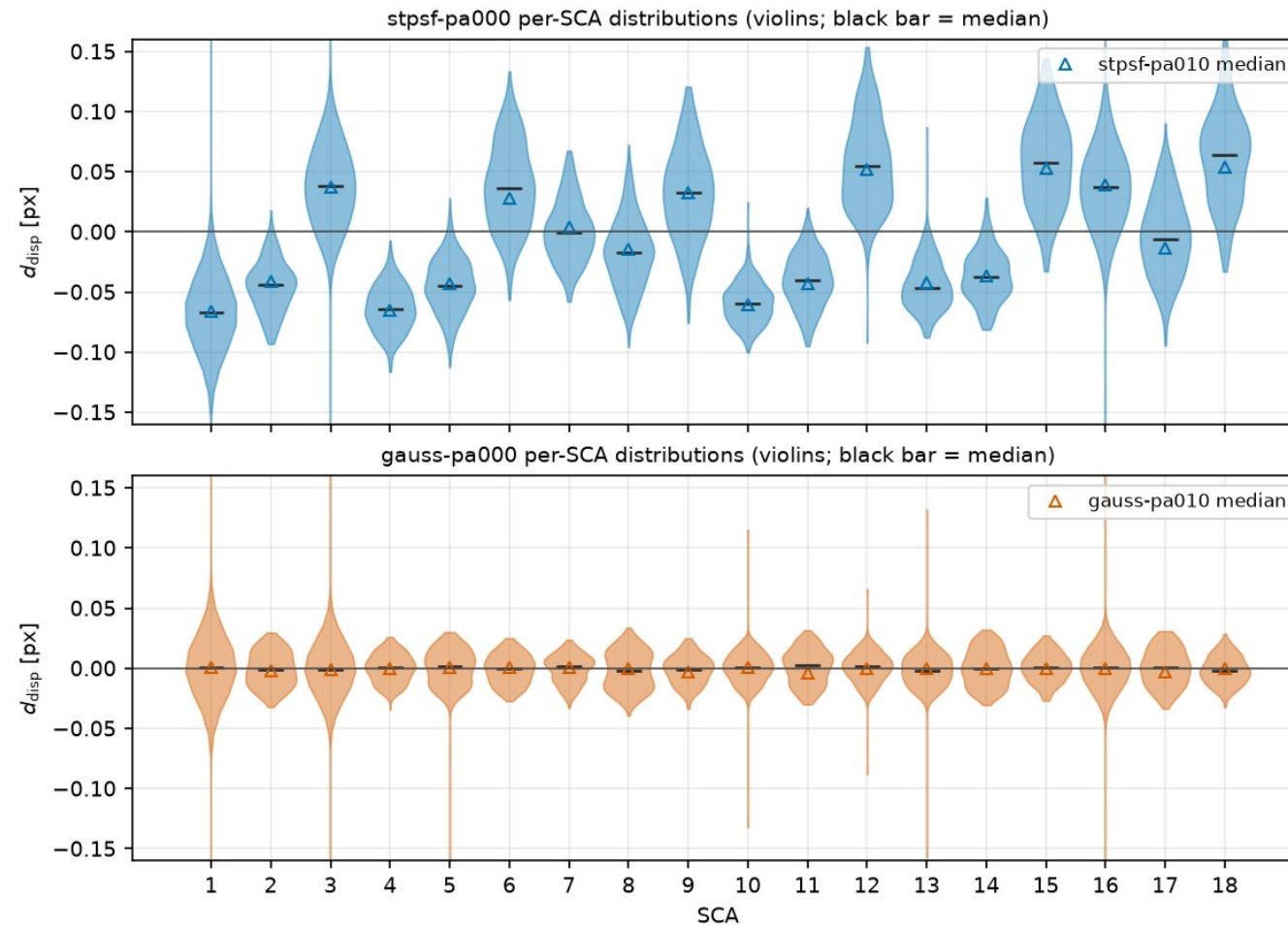
Every per-SCA median agrees with the 1× wave-1 run to $< 1\text{e-}5$ px, and with the 2026-07-03 validation run to ≤ 0.0036 px.

Results — per-SCA median offset



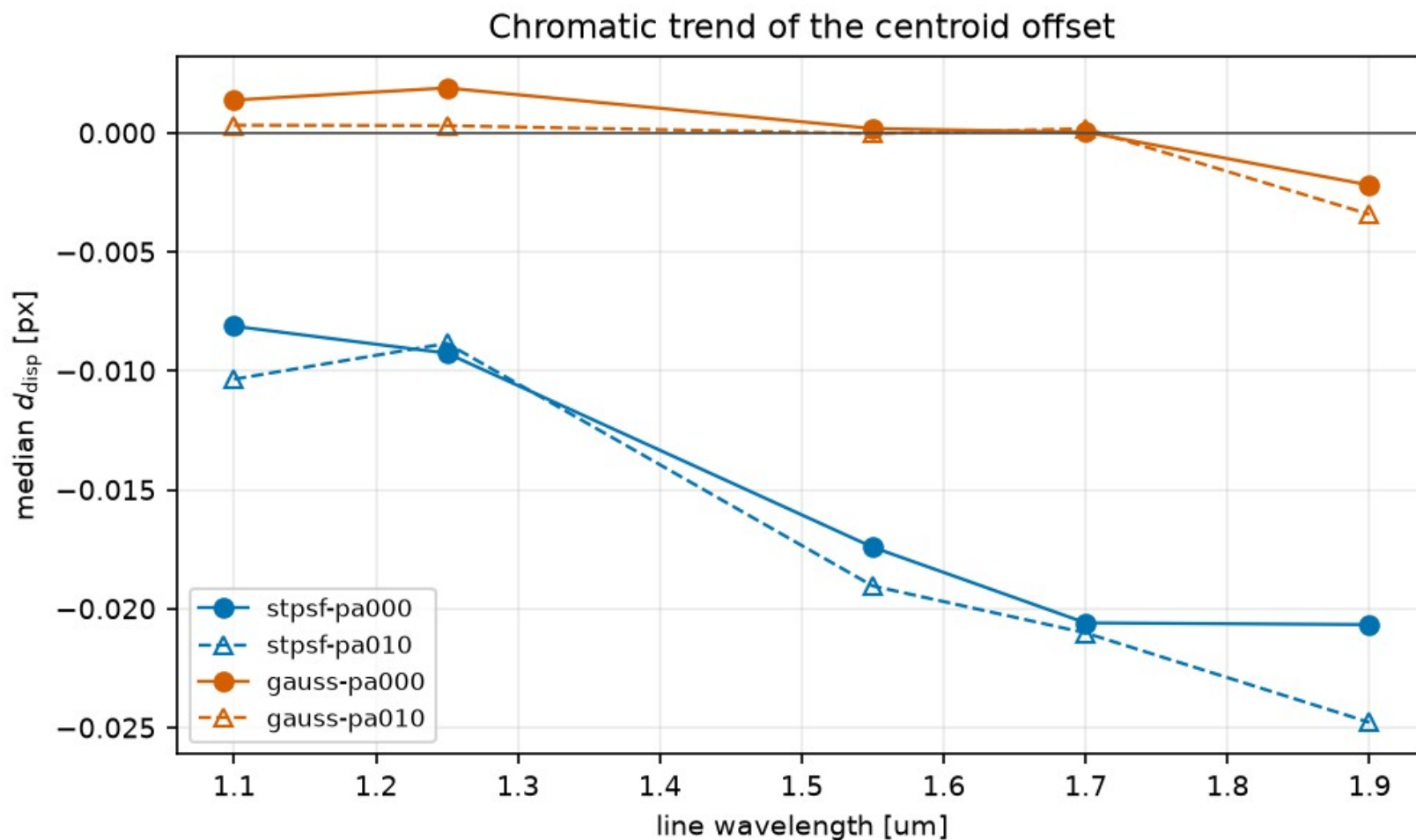
Median d_{disp} per SCA for the four runs, with the 2026-07-03 reference overlaid. STPSF runs show a fixed per-SCA pattern up to ± 0.065 px, the same at both PAs; Gaussian-control runs are consistent with zero on every SCA.

Results — per-SCA distributions



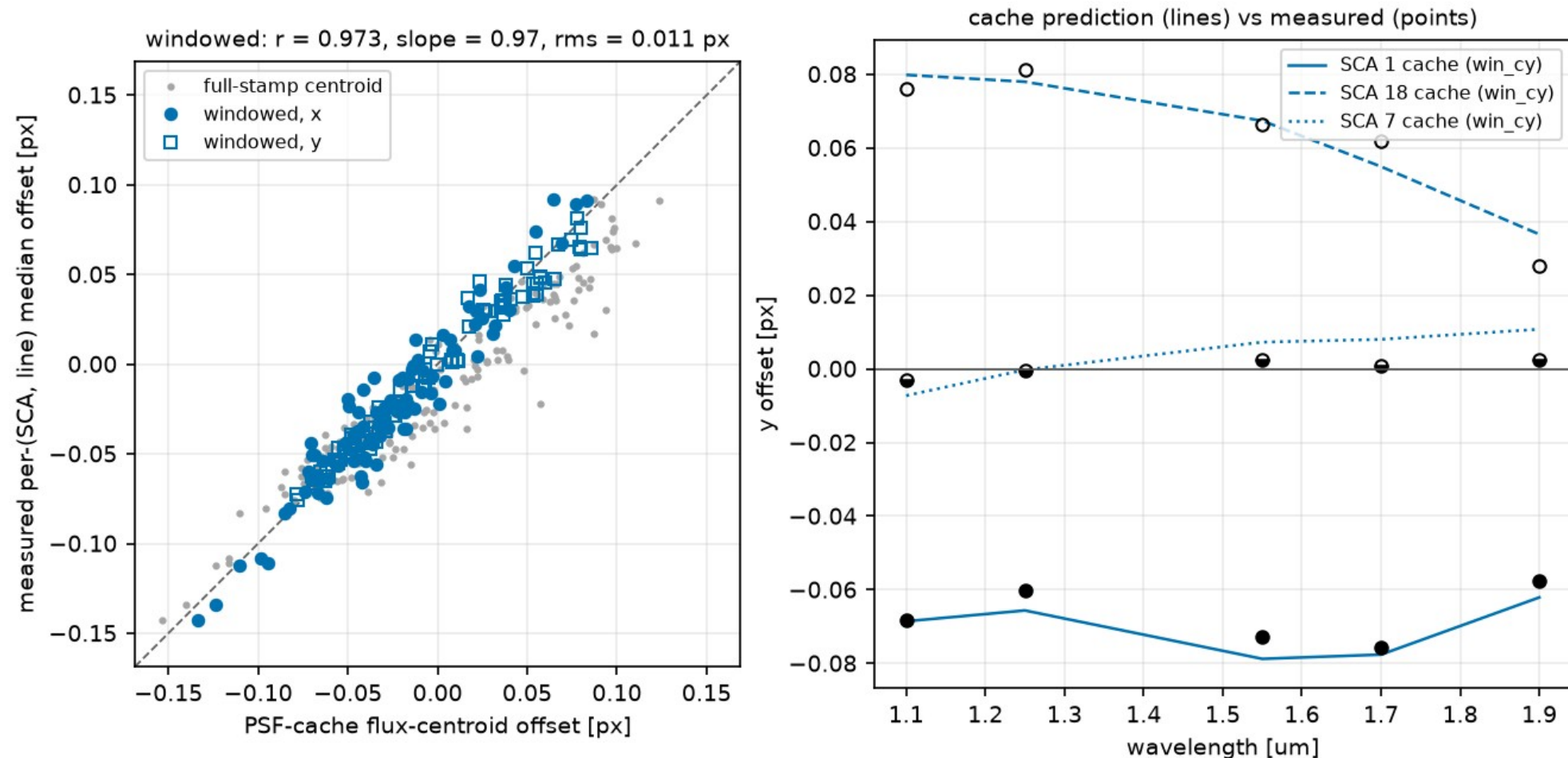
d_{disp} distributions per SCA (PA 0 runs; PA 10 medians as open triangles). Per-SCA MADs: 0.011–0.028 px (STPSF), 0.008–0.014 px (Gaussian).

Results — offset vs wavelength



Median d_{disp} per line, pooled over SCAs (opposite-sign SCAs partially cancel). STPSF runs trend from -0.008 px at $1.10\ \mu\text{m}$ to -0.021 px at $1.90\ \mu\text{m}$; Gaussian runs are flat within ± 0.003 px.

Results — comparison with PSF flux centroids



Measured per-(SCA, line) median offsets vs flux-weighted centroids computed directly from the PSF cache stamps with the same band/baseline/window procedure: $r = 0.973$, slope 0.97, rms 0.011 px. The Gaussian-cache centroids are zero to machine precision.

File guide — per-pointing contents

Inside each pointing directory (lines_{stpsf,gauss}/pointing_*/*):

<code>grism*_detSCA{NN}.fits</code>	per-SCA image: MODEL (counts/s) + ISIM (counts)
<code>grism*_detSCA{NN}.png</code>	quicklook PNG per SCA (+ one focal-plane mosaic)
<code>grism*_sources.parquet</code>	manifest of every (source, SCA, order) rendered
<code>grism*_meta.yaml</code>	pointing, exposure, RNG keys, batching, git SHA
<code>residuals.parquet</code>	measured vs predicted line positions (+1 order)
<code>truth_lines.parquet</code>	float64 truth positions, all orders and lines
<code>truth_lines_provenance.json</code>	writer script, commit, inputs, row counts

L2 trees (lines*_l2/pointing_*) hold one *_detSCA{NN}_l2.asdf per SCA.

catalog_lines/ holds the shared input catalog (next slides).

The following slides give the columns of each tabular file.

File: catalog_lines/metadata.parquet

One row per source (15,112 rows) — the simulation input catalog.

column	type	units	meaning
ra, dec	float64	deg	source position (ICRS)
type	str	—	"PSF" for all rows (point sources)
n, half_light_radius, pa, ba	float32	mixed	Sérsic morphology; trivial (0/0/0/1) for stars
F158	float32	maggies	apparent F158 flux, linear AB units; 3.0e-7 (mag 16.31) for every star
z_obs, z_cosmo	float32	—	redshifts; 0 for stars
sed_index	int32	—	row index into seds.zarr/star_seds
flux_scale	float32	maggies	SED multiplier applied at dispersal; equals F158 for stars
sim	int16	—	SED-storage partition; 0 for stars
src_index	int32	—	row in the parent reference catalog — cross-run provenance key

Files: catalog_lines/seds.zarr and lines.ecsv

seds.zarr — the line-only SED, f_λ (FLAM) on a vacuum-wavelength grid 9000–21000 Å in 2 Å steps (6001 samples); one shared template, referenced by sed_index. Scale by flux_scale for physical flux.

lines.ecsv — the injected line list:

column	units	meaning
line_id	—	0–4, blue to red; key used by residuals and truth tables
center_A	Å (vacuum)	11000 / 12500 / 15500 / 17000 / 19000
fwhm_A	Å	10.0 for all lines
amp_rel	—	line peak / local continuum at SED level: 5.0–10.0 (the continuum itself was removed; no_continuum = true)

provenance.json records the builder inputs: parent catalog, mag limit, continuum mag, line parameters, wavelength grid, build timestamp.

File: grism*_detSCA{NN}.fits

extension	content
0 PRIMARY	header only (metadata below)
1 MODEL	noise-free count-rate image, counts/s, float32, 4088×4088
2 ISIM	Poisson-sampled counts at 190.22 s, float32, 4088×4088

Primary-header keywords:

keyword	meaning
WFICENRA / WFICENDEC / WFICENPA	pointing RA, Dec, PA [deg]
DETNUM	SCA number 1-18
EXPTIME	exposure time [s]
MA_TABLE	MA table number (1036)
SEED / RNDSEED0 / RNDSEED1	top seed + per-SCA JAX RNG key words
GITSHA	pipeline git commit
PLAN, PASS, SEGMENT, OBS, VISIT, EXPOSURE	APT identifiers

File: grism*_sources.parquet

The dispersal manifest: one row per (source, SCA, order) actually rendered.

column	type	meaning
catalog_index	int64	row index into this run's catalog_lines/metadata.parquet
sca	int64	SCA number 1-18
order	str	spectral order: "0", "1", or "2"
type	str	"PSF" (all rows)
xsca, ysca	float64	undispersed source position on this SCA [1-indexed FITS px]
ra, dec	float64	source position [deg]
flux_scale	float64	SED multiplier applied at dispersal
F158	float64	F158 flux [maggies]; same on every row of a source

File: residuals.parquet

The closure measurement: one row per first-order (star, line) box — 8575 (PA 0) / 8465 (PA 10) rows.

column	type	meaning
sca	int64	SCA number
catalog_index	int64	source key (into catalog metadata)
xsca, ysca	float64	undispersed position [1-indexed FITS px]
line_id, center_A	int64/float64	which line (see lines.ecsv)
x_pred_jax, y_pred_jax	float32	JAX optical-model prediction [px]; float32 path, ULP $\sim 1e-3$ px
x_pred_class, y_pred_class	float64	reference-implementation prediction [px]
x_meas, y_meas	float64	measured centroid [px]; NaN if box off-detector
dx, dy	float32	$x_{\text{meas}} - x_{\text{pred_jax}}$, $y_{\text{meas}} - y_{\text{pred_jax}}$ [px]
d_disp, d_cross	float64	residual on local dispersion / cross-dispersion axes [px] — the science columns

File: truth_lines.parquet

The shipped truth table: one row per (source, SCA, order, line), orders 0/1/2 — 25,680 rows; positions evaluated in float64.

column	type	meaning
src_index	int32	row in the parent reference catalog — cross-run provenance key
catalog_index	int64	row in this run's catalog metadata
ra, dec	float64	source position [deg]
sca	int16	SCA number
order	str	"0", "1", "2"
xsca, ysca	float64	undispersed position [1-indexed FITS px]
line_id, center_A	int64/float64	which line [Å, vacuum]
fwhm_A, amp_rel	float64	line width and relative amplitude
x_pred, y_pred	float64	predicted dispersed line position [1-indexed FITS px]
on_detector	bool	prediction point inside $[0.5, 4088.5]^2$ (weaker than the residuals' box-fully-on-detector cut)

Files: L2 ASDF, meta YAML, audit records

*_detSCA{NN}_l2.asdf — romanisim L2 product per SCA (202 MB): roman.data / err / dq and var arrays (DN/s), full Roman metadata tree; MA table 1036, 11 resultants, effective exposure 190.22287 s.

grism*_meta.yaml — pointing, exposure, seed and per-SCA JAX RNG keys (reconstructable via `jax.random.wrap_key_data`), wavelength step, batching, per-SCA source counts by order, pipeline git SHA, APT identifiers.

scripts/ — verbatim copy, at submission time, of the campaign driver, configs, pointing tables and truth-table writer.

slurm-meta/ — per-job audit: the exact task script plus an .env record (job id, stage, partition, resources, config paths, git commits, timestamp).

logs/ — per-file romanisim wrap logs.